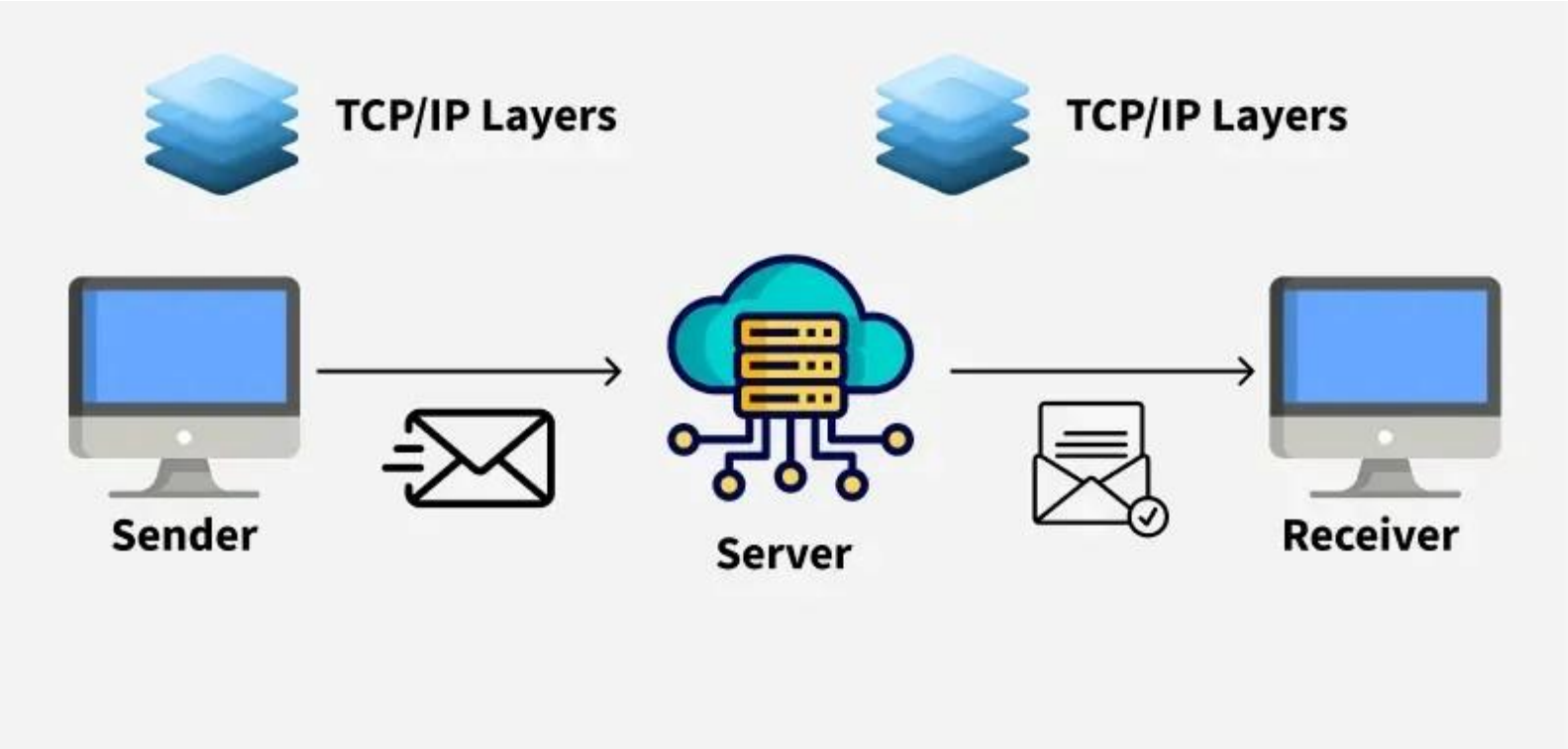
TCP/IP Model

The TCP/IP model is a framework that is used to model the communication in a network. It is mainly a collection of network protocols and organization of these protocols in different layers for modeling the network.

- It has four layers, Application, Transport, Network/Internet and Network Access.
- While the OSI model has seven layers, the 4 layer TCP/IP model is simpler and commonly used in today's Internet and networking systems.

Role of TCP/IP

One of its main goals is to make sure that the data sent by the sender arrives safely and correctly at the receiver's end. To do this, the data is broken down into smaller parts called packets before being sent. These packets travel separately and are reassembled in the correct order when they reach the destination.



TCP/IP

Note: *This helps prevent errors and makes sure the message is complete and accurate.*

Layers of TCP/IP Model

1. Application Layer

The Application Layer is the top layer of the TCP/IP model and the one closest to the user. This is where all the apps you use like web browsers, email clients, or file sharing tools connect to the network.

- It acts like a bridge between your software (like Chrome, Gmail, or WhatsApp) and the lower layers of the network that actually send and receive data.
- It supports different protocols like HTTP (for websites), FTP (for file transfers), SMTP (for emails), and DNS (for finding website addresses).
- It also manages things like data formatting, so both sender and receiver understand the data, encryption to keep data safe, and session management to keep track of ongoing connections.

2. Transport Layer

The Transport Layer is responsible for making sure that data is sent reliably and in the correct order between devices. It checks that the data you send like a message, file, or video arrives safely and completely.

- **This layer uses two main protocols:** TCP and UDP, depending on whether the communication needs to be reliable or faster.
- TCP is used when data must be correct and complete, like when loading a web page or downloading a file.
- It checks for errors, resends missing pieces, and keeps everything in order. On the other hand, UDP (User Datagram Protocol) is faster but doesn't guarantee delivery useful for things like live video or online games where speed matters more than perfect accuracy.

3. Internet Layer

The Internet Layer is used for finding the best path for data to travel across different networks so it can reach the right destination. It works like a traffic controller, helping data packets move from one network to another until they reach the correct device.

- This layer uses the Internet Protocol (IP) to give every device a unique IP address, which helps identify where data should go.
- The main job of this layer is routing deciding the best way for data to travel.
- It also takes care of packet forwarding (moving data from one point to another), fragmentation (breaking large data into smaller parts), and addressing.

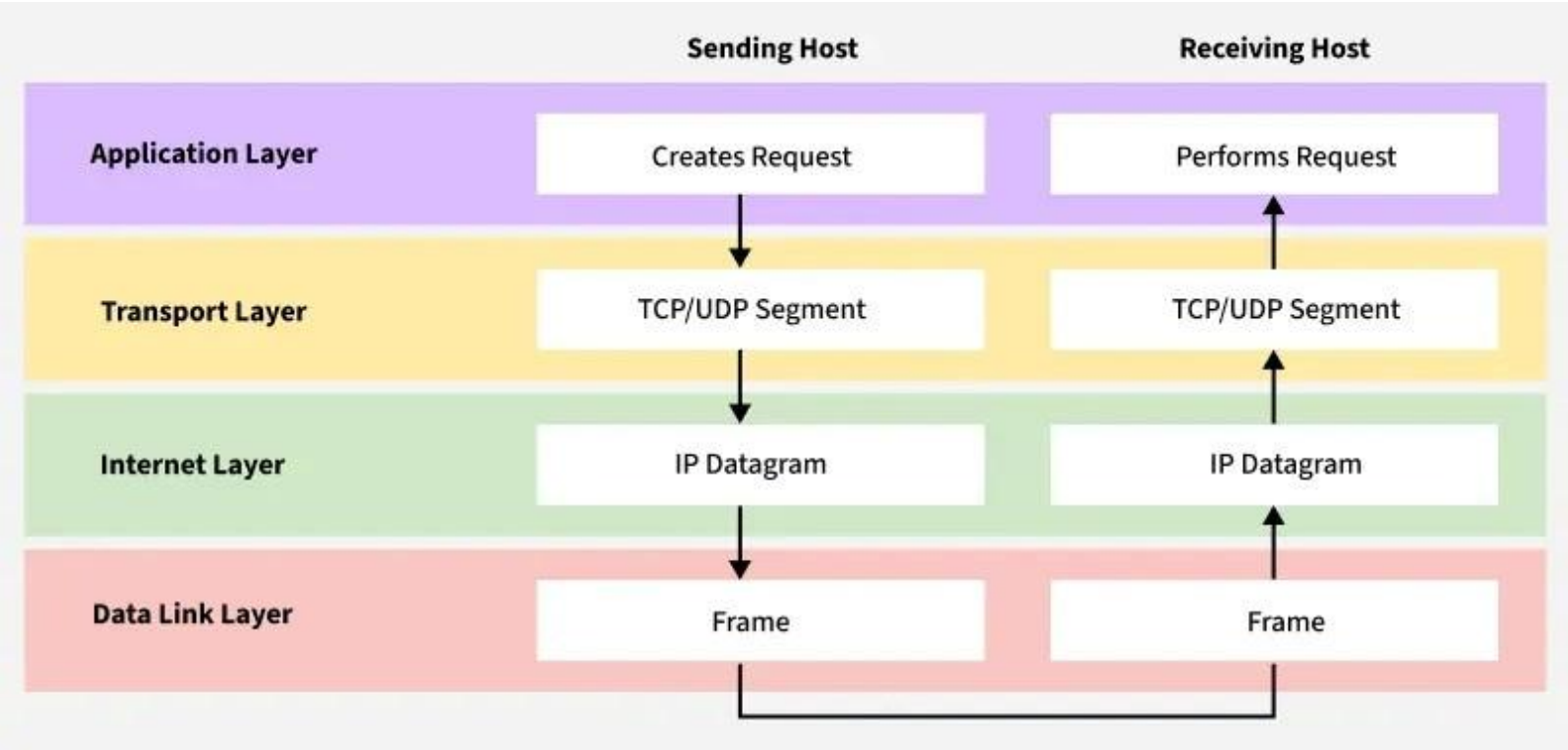
4. Network Access Layer

The Network Access Layer is the bottom layer of the TCP/IP model. It deals with the actual physical connection between devices on the same local network like computers connected by cables or communicating through Wi-Fi.

- This layer makes sure that data can travel over the hardware, such as wires, switches, or wireless signals.
- It also handles important tasks like using MAC addresses to identify devices, creating frames (the format used to send data over the physical link), and checking for basic errors during transmission.

Working of TCP/IP Model

The working of TCP/IP can be explained with the help of the diagram given below and explained :



TCP/IP WORKING MODEL

When Sending Data (From Sender to Receiver)

- **Application Layer**: Prepares user data using protocols like HTTP, FTP, or SMTP.

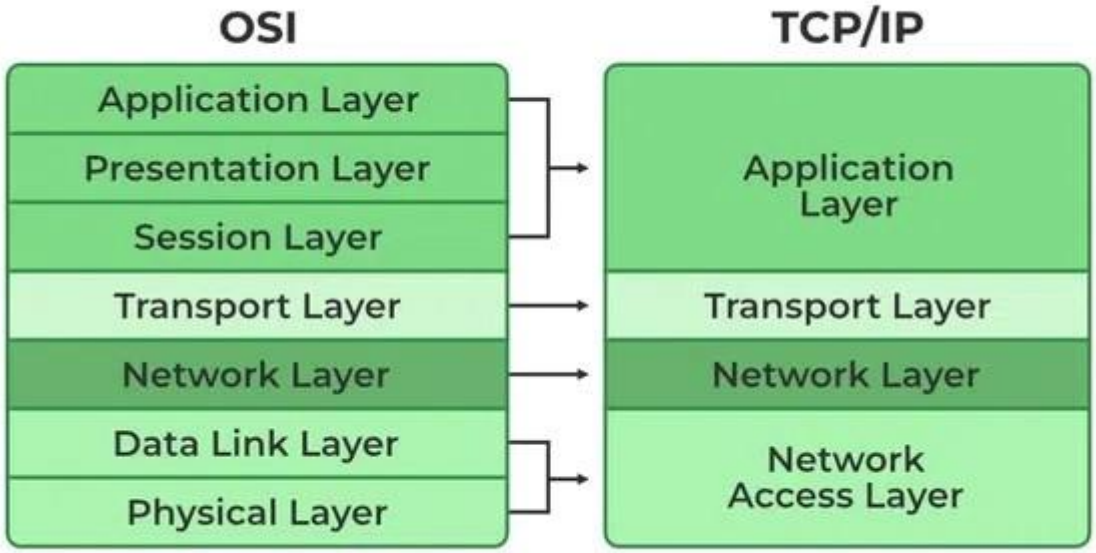
- **Transport Layer (TCP/UDP)**: Breaks data into segments and ensures reliable (TCP) or fast (UDP) delivery.
- **Internet Layer (IP)**: Adds IP addresses and decides the best route for each packet.
- **Link Layer (Network Access Layer)**: Converts packets into frames and sends them over the physical network.

When Receiving Data (At the Destination)

- **Link Layer**: Receives bits from the network and rebuilds frames to pass to the next layer.
- **Internet Layer**: Checks the IP address, removes the IP header, and forwards data to the Transport Layer.
- **Transport Layer**: Reassembles segments, checks for errors, and ensures data is complete.
- **Application Layer**: Delivers the final data to the correct application (e.g., displays a web page in the browser).

Why TCP/IP is Used Over the OSI Model

TCP/IP is used over the OSI model because it is simpler, practical, and widely adopted for real-world networking and the internet. The diagram below shows the comparison of OSI layer with the TCP:



OSI v/s TCP/IP

Reason	Explanation

Reason	Explanation
Simpler Structure	TCP/IP has only 4 layers, compared to 7 in OSI, making it easier to implement and understand in real systems.
Protocol-Driven Design	TCP/IP was designed based on working protocols, while the OSI model is more of a theoretical framework.
Flexibility and Robustness	TCP/IP adapts well to different hardware and networks and includes error handling, routing, and congestion control.
Open Standard	TCP/IP is open, free to use, and not controlled by any single organization, helping it gain universal acceptance.
Actual Use vs Conceptual Model	The OSI model is great for education and design principles, but TCP/IP is the one actually used in real-world networking.

Advantages of TCP/IP Model

- **Interoperability** : The TCP/IP model allows different types of computers and networks to communicate with each other, promoting compatibility and cooperation among diverse systems.
- **Scalability** : TCP/IP is highly scalable, making it suitable for both small and large networks, from local area networks (LANs) to wide area networks (WANs) like the internet.
- **Standardization** : It is based on open standards and protocols, ensuring that different devices and software can work together without compatibility issues.
- **Flexibility** : The model supports various routing protocols, data types, and communication methods, making it adaptable to different networking needs.
- **Reliability** : TCP/IP includes error-checking and retransmission features that ensure reliable data transfer, even over long distances and through various network conditions.

Disadvantages of TCP/IP Model

- **Security Concerns** : TCP/IP was not originally designed with security in mind. While there are now many security protocols available (such as SSL/TLS), they have been added on top of the basic TCP/IP model, which can lead to vulnerabilities.

- **Inefficiency for Small Networks** : For very small networks, the overhead and complexity of the TCP/IP model may be unnecessary and inefficient compared to simpler networking protocols.
- **Limited by Address Space** : Although IPv6 addresses this issue, the older IPv4 system has a limited address space, which can lead to issues with address exhaustion in larger networks.
- **Data Overhead** : TCP the transport protocol, includes a significant amount of overhead to ensure reliable transmission.

Types of Network Protocols and Their Uses

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Network protocols are a set of rules that are responsible for the communication of data between various devices in the network. These protocols define guidelines and conventions for transmitting and receiving data, ensuring efficient and reliable data communication.

What is Network Protocol?

A network protocol is a set of rules that govern data communication between different devices in the network. It determines what is being communicated, how it is being communicated, and when it is being communicated. It permits connected devices to communicate with each other, irrespective of internal and structural differences.

How do Network Protocols Work?

It is essential to understand how devices communicate over a network by recognizing network protocols. The Open Systems Interconnection (OSI), the most widely used model, illustrates how computer systems interact with one another over a network. The communication mechanism between two network devices is shown by seven different layers in the OSI model. Every layer in the OSI model works based on different network protocols. At every layer, one or more protocols are there for network communication. To enable network-to-network connections, the Internet Protocol (IP), for instance, routes data by controlling information like the source and destination addresses of data packets. It is known as a network layer protocol.

Types of Network Protocols

In most cases, communication across a network like the Internet uses the OSI model. The OSI model has a total of seven layers. Secured connections, network management, and network communication are the three main tasks that the network protocol performs. The purpose of protocols is to link different devices.

The protocols can be broadly classified into three major categories:

- Network Communication
- Network Management
- Network Security

1. Network Communication

Communication protocols are really important for the functioning of a network. They are so crucial that it is not possible to have computer networks without them. These protocols formally set out the rules and formats through which data is transferred. These protocols handle syntax, semantics, error detection, synchronization, and authentication. Below mentioned are some network communication protocol:

Hypertext Transfer Protocol(HTTP)

It is a layer 7 protocol that is designed for transferring a hypertext between two or more systems. HTTP works on a client-server model, most of the data sharing over the web is done through using HTTP.

Transmission Control Protocol(TCP)

TCP layouts a reliable stream delivery by using sequenced acknowledgment. It is a connection-oriented protocol i.e., it establishes a connection between applications before sending any data. It is used for communicating over a network. It has many applications such as emails, FTP, streaming media, etc.

User Datagram Protocol(UDP)

It is a connectionless protocol that lay-out a basic but unreliable message service. It adds no flow control, reliability, or error-recovery functions. UPD is functional in cases where reliability is not required. It is used when we want faster transmission, for multicasting and broadcasting connections, etc.

Border Gateway Protocol(BGP)

BGP is a routing protocol that controls how packets pass through the router in an independent system one or more networks run by a single organization and connect to different networks. It connects the endpoints of a LAN with other LANs and it also connects endpoints in different LANs to one another.

Address Resolution Protocol(ARP)

ARP is a protocol that helps in mapping logical addresses to the physical addresses acknowledged in a local network. For mapping and maintaining a correlation between these logical and physical addresses a table known as ARP cache is used.

Internet Protocol(IP)

It is a protocol through which data is sent from one host to another over the internet. It is used for addressing and routing data packets so that they can reach their destination.

Dynamic Host Configuration Protocol(DHCP)

it's a protocol for network management and it's used for the method of automating the process of configuring devices on IP networks. A DHCP server automatically assigns an IP address and various other configurational changes to devices on a network so they can communicate with other IP networks. it also allows devices to use various services such as NTP, DNS, or any other protocol based on TCP or UDP.

2. Network Management

These protocols assist in describing the procedures and policies that are used in monitoring, maintaining, and managing the computer network. These protocols also help in communicating these requirements across the network to ensure stable communication. Network management protocols can also be used for troubleshooting connections between a host and a client.

Internet Control Message Protocol(ICMP)

It is a layer 3 protocol that is used by network devices to forward operational information and error messages. ICMP is used for reporting congestions, network errors, diagnostic purposes, and timeouts.

Simple Network Management Protocol(SNMP)

It is a layer 7 protocol that is used for managing nodes on an IP network. There are three main components in the SNMP protocol i.e., SNMP agent, SNMP manager, and managed device. SNMP agent has the local knowledge of management details, it translates those details into a form that is compatible with the SNMP manager. The manager presents data acquired from SNMP agents, thus helping in monitoring network glitches, and network performance, and troubleshooting them.

Gopher

It is a type of file retrieval protocol that provides downloadable files with some description for easy management, retrieving, and searching of files. All the files are arranged on a remote computer in a stratified manner. Gopher is an old protocol and it is not much used nowadays.

File Transfer Protocol(FTP)

FTP is a Client/server protocol that is used for moving files to or from a host computer, it allows users to download files, programs, web pages, and other things that are available on other services.

Post Office Protocol(POP3)

It is a protocol that a local mail client uses to get email messages from a remote email server over a TCP/IP connection. Email servers hosted by ISPs also use the POP3 protocol to hold and receive emails intended for their users. Eventually, these users will use email client software to look at their mailbox on the remote server and to download their emails. After the email client downloads the emails, they are generally deleted from the servers.

Telnet

It is a protocol that allows the user to connect to a remote computer program and to use it i.e., it is designed for remote connectivity. Telnet creates a connection between a host machine and a remote endpoint to enable a remote session.

3. Network Security

These protocols secure the data in passage over a network. These protocols also determine how the network secures data from any unauthorized attempts to extract or review data. These protocols make sure that no unauthorized devices, users, or services can access the network data. Primarily, these protocols depend on encryption to secure data.

Secure Socket Layer(SSL)

It is a network security protocol mainly used for protecting sensitive data and securing internet connections. SSL allows both server-to-server and client-to-server communication. All the data transferred through SSL is encrypted thus stopping any unauthorized person from accessing it.

Hypertext Transfer Protocol(HTTPS)

It is the secured version of HTTP. this protocol ensures secure communication between two computers where one sends the request through the browser and the other fetches the data from the web server.

Transport Layer Security(TLS)

It is a security protocol designed for data security and privacy over the internet, its functionality is encryption, checking the integrity of data i.e., whether it has been tampered with or not, and authentication. It is generally used for encrypted communication between servers and web apps, like a web browser loading a website, it can also be used for encryption of messages, emails, and VoIP.

Some Other Protocols

Internet Message Access Protocol (IMAP)

- ICMP protocol is used to retrieve message from the mail server. By using ICMP mail user can view and manage mails on his system.

Session Initiation Protocol (SIP)

- SIP is used in video, voice, and messaging application. This protocol is used to initiating, Managing, Terminating the session between two users while they are communicating.

Real-Time Transport Protocol (RTP)

- This protocol is used to forward audio, video over IP network. This protocol is used with SIP protocol to send audio, video at real-time.

Rout Access Protocol (RAP)

- RAP is used in network management. It helps to user for accessing the nearest router for communication. RAP is less efficient as compared to SNMP.

Point To Point Tunnelling Protocol (PPTP)

- It is used to implement VPN (Virtual Private Network). PPTP protocol append PPP frame in IP datagram for transmission through IP based network.

Trivial File Transfer Protocol (TFTP)

- TFTP is the simplified version of FTP. TFTP is also used to transfer file over internet

Resource Location Protocol (RLP)

- RLP is used to assign the resource such as server, printer, or other devices over the internet to the user. It is used to locate the resource to the client for broadcast query.

<u>Domain Name System (DNS)</u>

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The Domain Name System (DNS) is a hierarchical and distributed naming system that translates human-readable domain names into IP addresses, enabling users to access websites easily.

- Converts domain names (e.g., www.geeksforgeeks.org) into IP addresses.
- Eliminates the need to remember numerical IP addresses.
- Works using a hierarchical structure of servers.
- Uses root, TLD, and authoritative DNS servers for resolution.
- Supports caching to improve speed and reduce network traffic.

Working of DNS

The DNS process can be broken down into several steps, ensuring that users can access websites by simply typing a domain name into their browser.

- **User Input:** You enter a website address (for example, www.geeksforgeeks.org) into your web browser.
- **Local Cache Check:** Your browser first checks its local cache to see if it has recently looked up the domain. If it finds the corresponding IP address, it uses that directly without querying external servers.
- **DNS Resolver Query:** If the IP address isn't in the local cache, your computer sends a request to a DNS resolver. The resolver is typically provided by your Internet Service Provider (ISP) or your network settings.
- **Root DNS Server:** The resolver sends the request to a root DNS server. The root server doesn't know the exact IP address for www.geeksforgeeks.org but knows which Top-Level Domain (TLD) server to query based on the domain's extension (e.g., .org).
- **TLD Server:** The TLD server for .org directs the resolver to the authoritative DNS server for geeksforgeeks.org.
- **Authoritative DNS Server:** This server holds the actual DNS records for geeksforgeeks.org, including the IP address of the website's server. It sends this IP address back to the resolver.
- **Final Response:** The DNS resolver sends the IP address to your computer, allowing it to connect to the website's server and load the pa**ge.**

Structure of DNS

DNS operates through a hierarchical structure, ensuring scalability and reliability across the global internet infrastructure. Here's how it's organized:

- **Root DNS Servers:** These are the highest-level DNS servers and know where to find the TLD servers. They are crucial for directing DNS queries to the correct locations.
- **TLD Servers:** These servers manage domain extensions like .com, .org, .net, .edu, .gov and others. They help route queries to the authoritative DNS servers for specific domains.
- **Authoritative DNS Servers:** These are the servers that store the actual DNS records for domain names. They are responsible for providing the correct IP addresses that allow users to reach websites.

Types of Domains

DNS helps manage a wide variety of domain types to organize the vast number of websites on the internet. Here are the primary categories:

- **Generic Domains:** These include top-level domains like .com, .org, .net and .edu. These are widely used and recognized across the world.
- **Country Code Domains:** These domains represent specific countries or regions, such as .in for India, .us for the United States, .uk for the United Kingdom and .jp for Japan.
- **Inverse Domains:** Used for reverse DNS lookups, these domains help map IP addresses back to domain names. Reverse DNS lookups are useful for diagnostics and security purposes, ensuring that the source of network traffic is legitimate. So DNS can provide both the mapping for example to find the IP addresses of geeksforgeeks.org then we have to type

Types of Domains

Domain Name Server

The client machine sends a request to the local name server, which, if the root does not find the address in its database, sends a request to the root name server, which in turn, will route the query to a top-level domain (TLD) or authoritative name server.

- The root name server can also contain some hostName to IP address mappings.
- The Top-level domain (TLD) server always knows who the authoritative name server is.
- So finally the IP address is returned to the local name server which in turn returns the IP address to the host.

DNS Lookup

DNS Lookup, also called DNS Resolution, is the process of translating a human-readable domain name into its corresponding IP address. The process involves:

- **DNS Resolver:** Initiates the lookup process. Also called a DNS client.

- **Recursive Query:** Resolver queries multiple servers on behalf of the client until the IP is found.
- **Iterative Query:** Resolver asks servers for the best answer available.
- **Non-Recursive Query:** Resolver queries a server that already has the record in its cache.

Types of DNS Queries

There are basically three types of DNS Queries that occur in DNS Lookup. These are stated below.

Types of DNS Queries

- **Recursive Query:** In this query, if the resolver is unable to find the record, in that case, DNS client wants the DNS Server will respond to the client in any way like with the requested source record or an error message.
- **Iterative Query:** Iterative Query is the query in which DNS Client wants the best answer possible from the DNS Server.
- **Non-Recursive Query:** Non-Recursive Query is the query that occurs when a DNS Resolver queries a DNS Server for some record that has access to it because of the record that exists in its cache.

DNS Caching and TTL (Time-to-Live)

- DNS caching is a mechanism that stores DNS records locally to avoid querying external DNS servers repeatedly for the same information. This speeds up the browsing experience and reduces network traffic.
- TTL (Time-to-Live) is the amount of time that a DNS record is cached before it expires. When the TTL expires, the cache is cleared and a fresh DNS query must be made. The TTL is defined by the authoritative DNS server, which can adjust this based on the nature of the domain.

Example: If the TTL for www.geeksforgeeks.org is 3600 seconds, the record is cached for 1 hour before a fresh lookup is needed.

DNS Security and DNSSEC (DNS Security Extensions)

Although DNS is essential for smooth internet navigation, it can be vulnerable to security threats. One common risk is DNS cache poisoning, where malicious actors inject harmful DNS records into caches, redirecting users to fraudulent websites.

DNS Security Extensions (DNSSEC)

- It is a protocol designed to address these security concerns.
- It adds cryptographic signatures to DNS records, allowing resolvers to verify the authenticity and integrity of DNS responses.
- DNSSEC ensures that the information a user receives from a DNS query has not been tampered with.

Reverse DNS Lookup

Reverse DNS Lookup is the process of mapping an IP address back to its corresponding domain name. This is the opposite of the typical DNS lookup, where a domain name is resolved to an IP address. Reverse DNS is commonly used for:

Reverse DNS Lookup

- **Network Diagnostics:** System administrators use reverse DNS to determine the domain name associated with a specific IP address. This helps identify the source of network traffic.
- **Email Security:** Many email servers perform reverse DNS lookups to verify that incoming emails are coming from legitimate sources. This helps prevent spam and fraud.

DNS Record Types (A, CNAME, MX, TXT, etc.)

DNS records are essential for defining how domain names are used and how services are configured. Here are some of the most commonly used DNS record types:

- **A Record:** This record maps a domain name to an IPv4 address (e.g., geeksforgeeks.org to 185.199.109.153). This is the most common DNS record used to point a domain to its website's IP address.
- **CNAME Record:** The Canonical Name (CNAME) record allows you to alias one domain name to another. For example, www.geeksforgeeks.org can be an alias for geeksforgeeks.org.
- **MX Record:** The Mail Exchange (MX) record defines which mail servers are responsible for receiving emails for a domain. This is crucial for setting up email services.
- **TXT Record:** The Text (TXT) record stores text-based information. It is commonly used to verify domain ownership and to implement email security protocols like SPF (Sender Policy Framework) and DKIM (DomainKeys Identified Mail).

CP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols used to interconnect network devices on the internet. TCP/IP is also used as a communications protocol in a private computer network -- an intranet or extranet.

The entire IP suite -- a set of rules and procedures -- is commonly referred to as TCP/IP. TCP and IP are the two main protocols, though others are included in the suite. The TCP/IP protocol suite functions as an abstraction layer between internet applications and the routing and switching fabric.

TCP/IP specifies how data is exchanged over the internet by providing end-to-end communications that identify how it should be broken into packets, addressed, transmitted, routed and received at the destination. TCP/IP requires little central management and is designed to make networks reliable with the ability to recover automatically from the failure of any device on the network.

Internet Protocol Version 4 (IPv4) is the primary version used on the internet today. However, due to a limited number of addresses, a newer protocol known as IPv6 was developed in 1998 by the Internet Engineering Task Force (IETF). IPv6 expands the pool of available addresses from IPv4 significantly and is progressively being embraced.

How are TCP and IP different?

The two main protocols in the IP suite serve specific functions and have numerous differences. The key differences between TCP and IP include the following:

TCP

- It ensures a reliable and orderly delivery of packets across networks.
- TCP is a higher-level smart communications protocol that still uses IP as a way to transport data packets, but it also connects computers, applications, web pages and web servers.
- TCP understands holistically the entire stream of data that these assets require to operate and it ensures the entire volume of data needed is sent the first time.
- TCP defines how applications can create channels of communication across a network.
- It manages how a message is assembled into smaller packets before they're transmitted over the internet and reassembled in the right order at the destination address.
- TCP operates at Layer 4, or the transport layer, of the Open Systems Interconnection (OSI model).
- TCP is a connection-oriented protocol, which means it establishes a connection between the sender and the receiver before delivering data to ensure reliable delivery.
- As it does its work, TCP can also control the size and flow rate of data. It ensures that networks are free of any congestion that could block the receipt of data. An example is an application that wants to send a large amount of data over the internet. If the application only used IP, the data would have to be broken into multiple IP packets. This would require multiple requests to send and receive data, as IP requests are issued per packet.
- With TCP, only a single request to send an entire data stream is needed; TCP handles the rest.
- TCP runs checks to ensure data is delivered. It can detect problems that arise in IP and request retransmission of any data packets that were lost.
- TCP can reorganize packets so they're transmitted in the proper order. This minimizes network congestion by preventing network bottlenecks caused by out-of-order packet delivery.

IP

- IP is a low-level internet protocol that facilitates data communications over the internet.
- IP delivers packets of data that consist of a header, which contains routing information, such as the source and destination of the data and the data payload itself.
- It defines how to address and route each packet to ensure it reaches the right destination. Each gateway computer on the network checks this IP address to determine where to forward the message.
- IP is limited by the amount of data it can send. The maximum size of a single IP data packet, which contains both the header and the data, is between 20 and 24 bytes. This means that longer strings of data must be broken into multiple data packets that have to be sent independently and then reorganized into the correct order.
- It provides the mechanism for delivering data from one network node to another.
- IP operates at Layer 3, or the network access layer, of the OSI model.
- IP is a connection-less protocol, which means it doesn't guarantee delivery nor does it provide error checking and correction.

Other components in a TCP/IP network

Other components present in a TCP/IP network include subnet masks, network address translation (NAT) and various protocols.

A subnet mask tells a computer, or other network device, what portion of the IP address is used to represent the network and what part is used to represent hosts, or other computers, on the network. A NAT is the virtualization of IP addresses. It helps improve security and decrease the number of IP addresses an organization needs.

Common TCP/IP protocols include the following:

- **Hypertext Transfer Protocol.** HTTP handles the communication between a web server and a web browser.
- **HTTP Secure.** HTTP Secure handles secure communication between a web server and a web browser.
- **File Transfer Protocol.** FTP handles transmission of files between computers.
- **Domain name system.** DNS translates domain names into IP addresses.
- **Simple mail transfer protocol.** SMTP is used for email communications and is responsible for the transmission of emails between mail servers.
- **User datagram protocols.** UDP is a connectionless protocol that offers faster but less dependable data delivery. It's widely used in real-time applications such as video streaming and online gaming.

How does TCP/IP work?

TCP/IP uses the client-server model of communication in which a user or machine -- a client -- is provided a service, such as sending a webpage, by another computer -- a server -- in the network.

Collectively, the TCP/IP suite of protocols is classified as stateless, which means each client request is considered new because it's unrelated to previous requests. Being stateless frees up network paths so they can be used continuously.

The transport layer itself, however, is stateful. It transmits a single message and its connection remains in place until all the packets in a message have been received and reassembled at the destination.

The TCP/IP model differs slightly from the seven-layer OSI networking model designed after it. The OSI reference model defines how applications can communicate over a network.

Why is TCP/IP important?

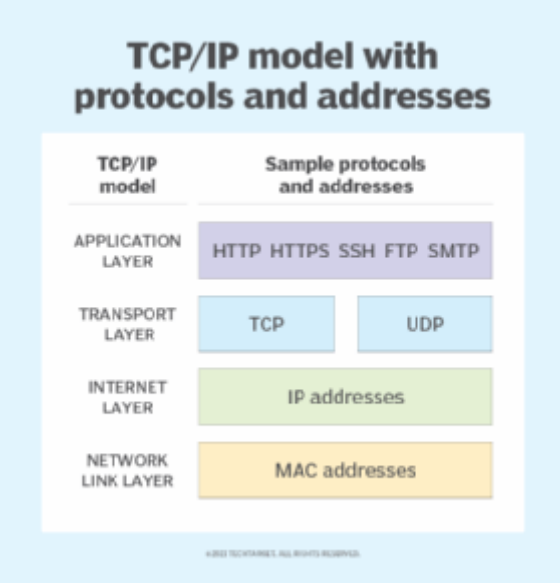
TCP/IP is the fundamental protocol suite that enables data transfer and communication across the internet and other networks. It's nonproprietary and, as a result, isn't controlled by any single company. Therefore, the IP suite can be modified easily. It's compatible with all operating systems (OSes), so it can communicate with any other system. The IP suite is also compatible with all types of computer hardware and networks.

TCP/IP is highly scalable and, as a routable protocol, can determine the most efficient path through the network. It's widely used in current internet architecture.

The 4 layers of the TCP/IP model

TCP/IP functionality is divided into the following four layers, each of which includes specific protocols:

1. **Application layer.** The application layer is the top layer and provides applications with standardized data exchange. Its protocols include HTTP, FTP, Post Office Protocol 3 (POP3), SMTP, DNS, Dynamic Host Configuration Protocol and SNMP. At the application layer, the payload is the actual application data.
2. **Transport layer.** The transport layer is responsible for maintaining end-to-end communications across the network. TCP handles communications between hosts and provides flow control, multiplexing and reliability. The transport protocols include TCP and User Datagram Protocol (UDP), which is sometimes used instead of TCP for special purposes.
3. **Internet layer.** The internet layer, also called the *network layer*, deals with packets and connects independent networks to transport the packets across network boundaries. The network layer protocols are IP and Internet Control Message Protocol, which are used for error reporting.
4. **Network link layer.** The network link layer, also known as the *network interface layer* or *data link layer*, consists of protocols that operate only on a link -- the network component that interconnects nodes or hosts in the network. The protocols in this lowest layer include Ethernet for local area networks and Address Resolution Protocol.



The four layers of the TCP/IP model.

Uses of TCP/IP

TCP/IP can be used for the following tasks:

- **Remote login and interactive file transfer.** TCP/IP provides remote login over the network for interactive file transfer to deliver email and webpages over the network.
- **Remote access to a file system.** TCP/IP provides remote access to a server host's file system, enabling users to access and manage files stored on the server from a remote location.
- **Represents information flow.** TCP/IP is used to represent how information changes form as it travels over a network from the concrete physical layer to the abstract application layer. It details the basic protocols, or methods of communication, at each layer as information passes through.
- **End-to-end data transmission.** It outlines how end-to-end communications should be achieved by dividing data into packets, addressing them, transmitting them, routing them and receiving them at the destination.

- **Cloud computing.** TCP/IP is used in cloud computing to facilitate communication between cloud-based services, applications and virtual machines. TCP/IP ensures that cloud resources and clients communicate reliably and securely across the internet.

Pros and cons of TCP/IP

The advantages of using the TCP/IP model include the following:

- It helps establish a connection between different types of computers.
- It works independently of the OS.
- TCP/IP supports many routing protocols.
- It uses a client-server architecture that's highly scalable.
- TCP/IP can be operated independently.
- It supports several routing protocols.
- It's lightweight and doesn't place unnecessary strain on a network or computer.

The disadvantages of TCP/IP include the following:

- It's complicated to set up and manage.
- The transport layer doesn't guarantee the delivery of packets.
- It isn't easy to replace protocols in TCP/IP.
- It doesn't clearly separate the concepts of services, interfaces and protocols, so it isn't suitable for describing new technologies in new networks.
- It's especially vulnerable to synchronization attacks, which are a type of denial-of-service attack in which a bad actor uses TCP/IP.

TCP/IP model vs. OSI model

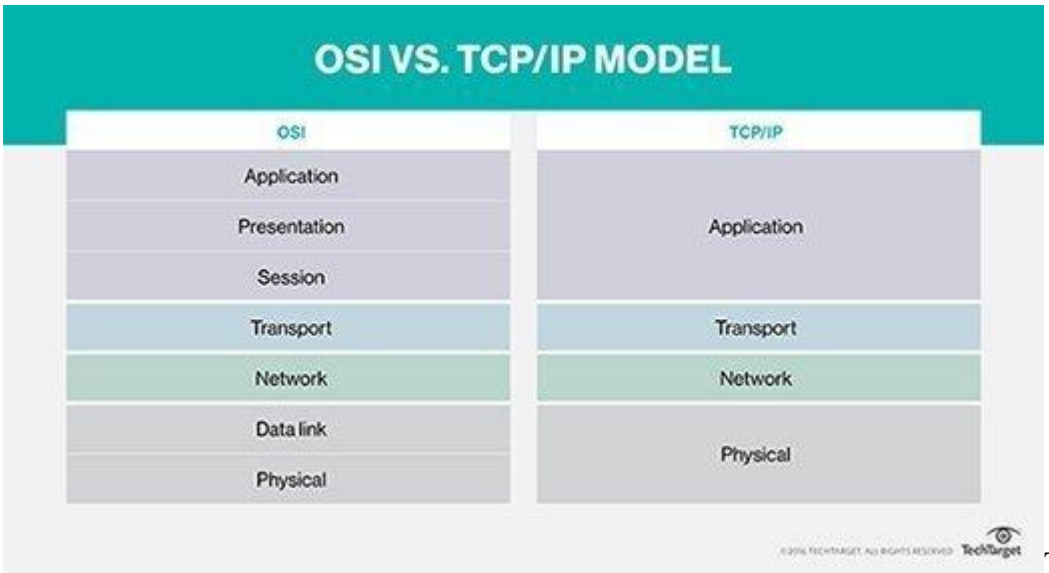
TCP/IP and OSI are the most widely used communication networking protocols. The main difference between the two models is that OSI is a conceptual model that isn't practically used for communication. Rather, it defines how applications can communicate over a network. TCP/IP, on the other hand, is a practical execution that's widely used to establish links and network interaction.

The TCP/IP protocols lay out standards on which the internet was created, while the OSI model provides guidelines on how communication must be done. Therefore, TCP/IP is a more practical model.

The TCP/IP and OSI models have similarities and differences. The main similarity is in the way they're constructed to both use layers, although TCP/IP consists of just four layers, while the OSI model consists of the following seven layers:

- **Layer 7:** the application layer, lets the user -- software or human -- interact with the application or network when the user wants to read messages, transfer files or engage in other network-related activities.
- **Layer 6:** the presentation layer**,** translates or formats data for the application layer based on the semantics or syntax that the app accepts.
- **Layer 5:** the session layer, sets up, coordinates and terminates conversations between apps.

- **Layer 4:** the transport layer, transfers data across a network and provides error-checking mechanisms and data flow controls.
- **Layer 3:** the network layer**,** moves data into and through other networks.
- **Layer 2**, the data link layer**,** handles problems that occur as a result of bit transmission errors.
- **Layer 1:** the physical layer, transports data using electrical, mechanical or procedural interfaces.



The differences between TCP/IP and OSI model.

The application layer is the upper layer for both the TCP/IP model and the OSI model. Although this layer performs the same tasks in each model, those tasks can vary depending on the data each receives.

The functions performed in each model are also similar because each uses a network layer and transport layer to operate. The TCP/IP and OSI models are each mostly used to transmit data packets. Although they use different means and different paths, they still reach their destinations.

The similarities between the TCP/IP model and the OSI model include the following:

- They're both logical models.
- They define networking standards.
- They divide the network communication process into layers.
- They provide frameworks for creating and executing networking standards and devices.
- They enable one manufacturer to make devices and network components that can coexist and work with the devices and components made by other manufacturers.

The differences between the TCP/IP model and the OSI model include the following:

- TCP/IP uses just one layer -- the application layer -- to define the functionalities of the upper layers, while OSI uses three layers -- application, presentation and session.
- TCP/IP uses one layer -- the physical layer -- to define the functionalities of the bottom layers, while OSI uses two layers -- physical and data link.
- The TCP/IP header size is 20 bytes, while the OSI header is 5 bytes.

- TCP/IP is a protocol-oriented standard, whereas OSI is a generic model based on the functionalities of each layer.
- TCP/IP follows a horizontal approach, while OSI follows a vertical approach.
- In TCP/IP, the protocols were developed first and then the model. In OSI, the model was developed first and then the protocols in each layer were developed.
- TCP/IP helps establish a connection between different types of computers, whereas OSI helps standardize routers, switches, motherboards and other hardware.